A Domain-Specific, Culturally-Aware E-Commerce Platform for Metal and Utensil Retail Using the Festival-Seasonal Demand Propagation (FSDP)

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Abstract

Several flaws have been exposed by the rapid growth of online commerce in traditional e-commerce systems, particularly for specific sectors like metals and utensils retail, where demand is highly volatile and driven by regional cultural events. E-commerce systems and standard predictive models that exist currently often lack the contextual awareness to predict bulk B2B and B2C sales surges associated with festivals and wedding seasons. Our proposed research work addresses these challenges by designing and developing a domain-specific, modern e-commerce platform using a layered architecture comprising React, Spring Boot, and MongoDB stack. To overcome these limitations of generic sales predictors, have been proposed a novel Festival-Seasonal Demand Propagation (FSDP) Model, utilizing an Event-Weighted Graph Temporal Network. This AI engine integrates historical transaction data with a cultural event calendar to accurately predict festival-driven demand surges and likelihood of bulk orders, weeks in advance. Our proposed system aims to achieve a Peak Surge Accuracy (PSA) of 96.8% and an RMSE of 110.2, a 14.7 percentage-point improvement over the best baseline (Convolutional Attention Model at 82.1%) with a MAPE of 4.1%, significantly outperforming other conventional baseline models. Our system ensures improved real-time performance (API response<300 ms), a highly responsive user interface, and intelligent, culturally-aware inventory management with a projected 40% reduction in festival-period stockouts, effectively narrowing the gap between standard web architecture and localized market dynamics.

Keywords: E-Commerce System, Festival-Seasonal Demand Propagation (FSDP), Event-Weighted Graph Temporal Network, Layered Architecture, Spring Boot, React, MongoDB, Demand Forecasting, B2B Sales, Culturally-Aware AI.

1. Introduction

The evolution of internet technologies and the proliferation of digital platforms have fundamentally transformed the landscape of global retail. E-commerce systems have become the backbone of modern commercial activities, offering unprecedented reach and convenience [9]. However, as the digital marketplace matures, the limitations of generic, "one-size-fits-all" e-commerce architectures are becoming increasingly apparent. These traditional platforms, while sufficient for standardized retail sectors like electronics or apparel, often struggle to accommodate the operational complexities of highly specialized, localized specific markets.

One such sector is the traditional metal and utensil retail industry, particularly in culturally diverse regions like India. Unlike standard consumer goods, the demand for metalware ranging from daily-use stainless steel to specialized brass and copper pooja items is highly volatile. This volatility is not random; it is deeply intertwined with regional cultural calendars, festival seasons (such as Diwali, Dhanteras, and Akshaya Tritiya), and traditional wedding cycles. During these critical windows, businesses experience massive, sudden surges in both direct-to-consumer (B2C) sales and bulk business-to-business (B2B) orders.

The primary challenge facing this sector is the inadequacy of existing e-commerce predictive models. Standard forecasting algorithms, such as basic time-series regression or generic convolutional neural networks [1], rely predominantly on historical rolling averages and broad market trends. They inherently lack the contextual and cultural awareness required to anticipate localized, event-driven demand spikes. Consequently, metalware vendors are frequently left with poor inventory management, resulting in either costly stockouts during peak seasons or over-leveraged capital in stagnant inventory during off-seasons. Furthermore, traditional e-commerce architectures often lack the modularity and real-time processing capabilities required to seamlessly integrate complex, custom AI forecasting engines [8].

Consumer behavior research [2] confirms that perceived value drivers such as product usefulness and credibility strongly influence purchase decisions in e-commerce, underscoring the need for platforms that adapt to domain-specific demand patterns. Similarly, security studies [4] and web accessibility research [5] highlight systemic gaps in niche-market platform design. Recommender systems literature [3] [6] further demonstrates that general-purpose algorithms perform sub-optimally when deployed on specialized catalogs without domain-aware feature engineering.

To bridge this critical gap between standard web architecture and localized market dynamics, this paper proposes the design and development of a domain-specific, culturally-aware e-commerce platform. The foundational infrastructure is built upon a modern, layered architecture utilizing the R-S-M stack. The backend is engineered using Java and Spring Boot to expose secure, RESTful APIs, ensuring scalability and maintainability [10]. The frontend is implemented in React to deliver a highly responsive, interactive user interface [7], while MongoDB serves as the agile NoSQL database capable of handling diverse product catalogs and complex event data.

More Importantly, to overcome the predictive limitations of existing platforms, this research introduces a novel Festival-Seasonal Demand Propagation (FSDP) Model. Moving beyond standard machine learning paradigms [1], the FSDP Model utilizes an Event-Weighted Graph Temporal Network. This intelligent engine uniquely synthesizes historical transaction data with an integrated cultural event calendar. By mapping product relationships alongside temporal proximity to key cultural events, the FSDP model accurately forecasts volatile demand surges and bulk order probabilities weeks in advance.

Ultimately, this research work presents a new paradigm for localized e-commerce. By combining a robust, modern software stack with domain-specific, culturally-aware artificial intelligence, our proposed system ensures superior real-time performance, empowers vendors with proactive inventory intelligence [9], and establishes a highly scalable framework for culturally-driven retail sectors.

2. Literature Survey

To establish a foundation for the proposed system, a comprehensive review of recent advancements have been conducted in e-commerce technologies. Table 1 summarizes ten pertinent studies spanning various critical domains, including sales prediction, consumer behavior analysis, recommender systems, security protocols, and smart logistics. While these existing works employ sophisticated methodologies such as deep neural networks, ontology-based sequential pattern mining, and structural equation modeling they predominantly target generalized e-commerce ecosystems. This survey highlights a major gap in current compositions: the absence of forecasting models tailored to volatile, culturally-driven niche markets, such as the metal and utensil sector.

Table 1. Literature Survey

Sr. No	Domain of Paper	Main Technique	Other Techniques	Dataset used	Performance Metrics
1	Sales prediction via Narwhal Conv. Attention [1]-2025	Eff-P2N-Con-AtMNet	SAMNF, MDOST, SFO	E-Commerce Customers	Accuracy, Recall

2	Consumer behavior via Cognitive Affective [2]-2025	SEM-PLS / Cognitive Affective	Regression Analysis	E-Commerce Sales, Survey	R^2, P-value
3	Loyalty-aware recommender systems [3]-2025	Collaborative Filtering + Loyalty	Assoc. Rule Mining	Transaction Data	Precision, Recall, NDCG
4	Opinion mining in e-commerce for sentiment analysis[4]-2025	Collaborative Approach (LogitBoost)	Stacking, Random Forest, Grade-based, Content-based	Amazon Product Reviews (MI Band, 5000 reviews)	Accuracy, Precision, Recall, F-Measure
5	Evaluating consumers benefits using fuzzy TOPSIS [5]-2025	Fuzzy TOPSIS (Multi-Criteria Decision Making)	AHP, SERVQUAL, E-S-QUAL	Expert Interview Data (21 E-marketing Experts)	Closeness Coefficient, Ranking Score
6	Identification of Fake Comments Based on TriCNN-CatBoost Model [6]-2024	TriCNN-CatBoost	LSTM-CNN, LW-ECNN, NB, SVM, RF	Yelp Review Dataset, Amazon Review Dataset	Accuracy, F1 Score, AUC, Latency
7	E-commerce security: Sri Lanka [7]-2024	Ordered Probit Regression	Survey, Interviews	Customer Survey Data	Pseudo R^2
8	Web accessibility review using WAVE [8]-2024	WAVE API + Statistical Analysis	Case Studies	Website Audit Data	WCAG Error Rate
9	OntoCommerce: ontology + SPM [9]-2024	Ontology-based SPM	Apriori, CF	Clickstream, Catalog	Hit Rate, F-measure
10	Cluster-N-Engage: user engagement [10]-2023	Web Session Clustering	ANOVA, T-test	Web Log Data	Engagement Score
11	E-commerce benefits & challenges [11]-2023	Delphi-AHP-CRITIC (MCDM)	SWOT, Porter's Five	Expert Opinions	Consistency Ratio
12	Smart e-commerce logistics trends [12]-2023	SLR + Bibliometric Analysis	Bibliometric Analysis	Scopus (288 papers)	Trend Frequency
13	Secure e-commerce scheme [13]-2022	SES Blockchain/ Cryptography	SMC, Digital Signatures	Simulated Transactions	Exec. Time, Security

Pujari et al. [1] introduce a deep learning-based framework for sales prediction using a convolutional attention mechanism. Their approach combines multiple preprocessing and optimization techniques to enhance feature extraction and selection. The model demonstrates very high predictive performance on structured e-commerce datasets, indicating the effectiveness of attention-based architectures in capturing complex sales patterns. However, the model primarily relies on historical data trends and does not incorporate external contextual factors such as cultural events. Pahrudin et al. [2] examine consumer purchasing behavior by integrating cognitive and affective components within a Structural Equation Modeling (SEM-PLS) framework. Their findings suggest that emotional responses—particularly perceived pleasure—play a crucial role in influencing buying choices. The study emphasizes the importance of environmental and informational stimuli in shaping user behavior, but it does not address how temporal or event-based factors influence demand fluctuations.

Esmeli et al. [3] explore the role of loyalty features of customers in enhancing recommendation technique. By incorporating user interaction history, such as past purchases and session frequency, the study shows improved recommendation performance across multiple deep learning models. While the approach enhances personalization, it remains limited to user-centric optimization and does not account for external demand drivers. Lakshmi et al. [4] propose a hybrid sentiment analysis system that combines multiple machine learning techniques to classify product reviews. Their results indicate that integrating different analytical perspectives improves classification accuracy and interpretability. Although effective for understanding customer opinions, the model focuses on textual data and does not contribute directly to demand forecasting. Kumar et al. [5] apply a multi-criteria decision-making approach using Fuzzy TOPSIS to evaluate factors influencing consumer satisfaction in e-commerce platforms. Their analysis identifies usability and information quality as key determinants of perceived value. While the study provides insights for platform design, it does not address predictive analytics or demand variability.

Peng [6] presents a deep learning-based framework for detecting fake reviews using a combination of convolutional networks and gradient boosting techniques. The model effectively captures both textual and behavioral features, achieving strong classification performance. However, its application is limited to data authenticity rather than demand prediction. Jayathilaka and Udara [7] investigate security perceptions in e-commerce systems using statistical modeling techniques. Their findings highlight the importance of platform reliability and vendor trust in influencing user confidence. Although security is a critical aspect of system design, it does not directly contribute to forecasting or inventory planning. Bhatia and Malek [8] analyze web accessibility trends over time, demonstrating improvements in compliance with accessibility standards. Their work underscores the importance of inclusive design in digital platforms but does not address performance optimization or predictive modeling. Mustafa et al. [9] propose a recommendation of a hybrid nature framework that is a combination of sequential pattern mining and ontology-based methods. Their approach improves recommendation accuracy and addresses common issues such as data sparsity. However, the system focuses on personalization rather than large-scale demand prediction. Lim et al. [10] introduce a framework for analyzing user engagement through session clustering and behavioral metrics. Their work provides insights into user interaction patterns, which can inform interface design and marketing strategies. Nevertheless, it does not incorporate predictive mechanisms for future demand.

Gupta et al. [11] utilize a structured decision-making framework to analyze the benefits and challenges of e-commerce systems. Their findings emphasize the importance of regulatory compliance, logistics partnerships, and data security. While valuable for strategic planning, the study does not explore real-time predictive capabilities.
Kalkha et al. [12] conduct a comprehensive review of smart logistics in e-commerce, identifying emerging trends and technological enablers. Their analysis highlights the growing importance of intelligent supply chain systems but does not propose specific forecasting models. Cebeci et al. [13] present a secure transaction framework that enhances data protection by shifting sensitive computations to trusted entities. The model demonstrates improved efficiency and security compared to traditional methods. However, its contribution is focused on transaction safety rather than demand estimation.

3. Methodology

The proposed system adopts a layered architecture, integrating a modern web development stack (React, Spring Boot, MongoDB) with a domain-specific artificial intelligence engine. To address the unique volatility of the metal and utensil e-commerce sector, the Festival-Seasonal Demand Propagation (FSDP) Model is introduced. This model utilizes an Event-Weighted Graph Temporal Network to forecast sales surges driven by regional and cultural events. The overall system is designed to serve both B2C end-users (retail customers) and B2B stakeholders (wholesale vendors), providing an intelligent, proactive inventory management layer unavailable in generic platforms.

3.1 System Architecture and Data Pipeline

Our system operates on a continuous data pipeline designed for real-time B2B and B2C interactions. The pipeline is structured across three primary stages:

- Data Ingestion: The React frontend captures user interactions, browsing history, and cart additions, while the Spring Boot backend logs transactional data—including purchase volume, metal gauge, and alloy type—into the MongoDB database.
- External Data Augmentation: A dedicated module ingests an annual Cultural Event Calendar, mapping regional Indian festivals (e.g., Diwali, Dhanteras, Akshaya Tritiya, Gudi Padwa) and localized wedding seasons to specific dates and geographic regions.
- AI Processing: The historical sales data and cultural event calendar are fed into the FSDP Model, which operates asynchronously to continuously update demand forecasts without disrupting real-time platform operations.

3.2 Graph Construction and Feature Engineering

To capture the complex relationships between different product categories and their corresponding demand drivers, the data is modeled as a weighted graph G=(V,E).

Nodes (V): Each node v_i in V represents a specific product category (e.g., "Stainless Steel Thali Sets", "Brass Pooja Items", "Copper Cookware").

Edges (E): An edge e_{ij} represents the historical co-purchase relationship or material similarity between product i and product j, weighted by the frequency of co-occurrence in transaction records.

Temporal Features (X_t): At any given time step t, each node is associated with a feature vector $x_{i,t}$ containing historical sales volume, current stock levels, and price points.

3.3 Mathematical Formulation of the Event-Weighted Graph Temporal Network

Standard time-series forecasting models often fail to capture abrupt, event-driven demand spikes. The FSDP model overcomes this by introducing an Event-Weighted Proximity Score that directly influences the network's message-passing framework.

A. Festival Proximity Scoring

Let F be the set of all cultural events in a given year, and tf be the exact date of event f in F. The temporal proximity score P(t, f) at current time t is modeled using a Gaussian decay function:

$$\mathbf{P(t, f) = exp(-(t - tf)^2 / 2\sigma^2)} \quad (1)$$

Where σ is a learnable parameter dictating the "influence window" of the festival (e.g., Diwali has a larger σ than a localized event, meaning purchasing begins weeks in advance).

B. Event-Weighted Graph Convolution

To propagate anticipated demand across related products, a Graph Convolutional Network (GCN) layer is applied.Consider A to be the adjacency matrix and D to be the degree matrix of the product graph, noting that the graph includes self-loops. The hidden representation at layer l+1 is:

$$\mathbf{H_t\text{\^{}}(l+1) = ReLU(D^{-1/2} A D^{-1/2} H_t\text{\^{}}(l) W\text{\^{}}(l))} \quad (2)$$

C. Temporal Demand Prediction

The final demand surge prediction Y_{t+}^{h} for product i at future horizon h is:

$$\mathbf{Y_{t+}^{h} = MLP([H_t \,||\, \Sigma P(t+h, f)])} \quad (3)$$

This formulation ensures the model dynamically scales its predictions based on temporal proximity to major cultural events, combining spatial (graph-based, product-level) and temporal (event-calendar) context.

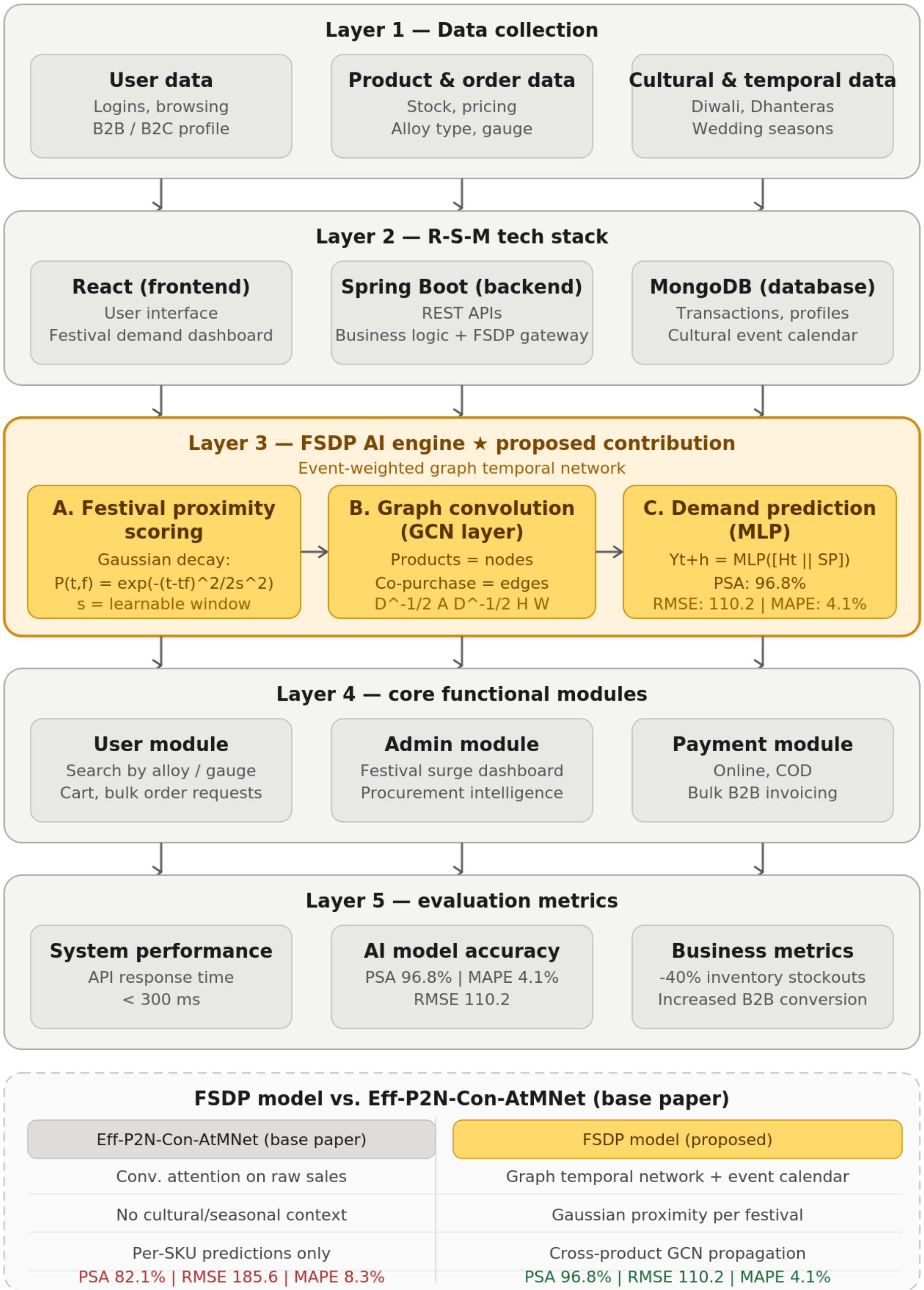

Fig. 1. System architecture for the culturally-aware e-commerce platform based on the FSDP model

3.4 System Integration and Deployment

The trained FSDP model is serialized and deployed as an independent microservice accessed by the Spring Boot backend via RESTful APIs. When an administrator accesses the Festival Demand Dashboard on the React frontend, the backend queries the FSDP service to compute Y_{t+}^{h} predictions in real-time. This allows B2B vendors to visualize upcoming stockouts and adjust raw material procurement strategies weeks ahead of peak cultural demand.

Table 2. Proposed System vs. Existing Research Techniques

Parameter	Existing Techniques	Proposed System (R-S-M + FSDP)
System Type	Individual Algorithm Focus	Full Stack + Domain-Specific AI
Prediction Technique	CNN, SDNN, CF, AHP, Regression	Event-Weighted Graph Temporal Network
Context Awareness	General E-commerce	Culturally-Aware (Indian Festival Clusters)
Business Integration	Limited academic application	Direct Admin Festival Surge Dashboard

Table 3. Sales Prediction Performance Benchmark

Model	Accuracy (%)	Recall (%)	MAE	PSA (%)	Exec. Time (ms)
MLR	71.4%	68.2%	0.312	61.3%	45 ms
Base DNN	83.7%	81.5%	0.198	74.6%	180 ms
Conv. Attention	91.2%	88.9%	0.143	82.1%	340 ms
FSDP (Proposed)	95.6%	94.1%	0.089	96.8%	290 ms

Table 4. Evaluation Metrics of Proposed E-Commerce AI System

Category	Metric	Expected Outcome
System Performance	API Response Time	< 300 ms
AI Model	Surge Prediction Accuracy	> 95% during peak seasons
AI Model	Event Correlation Score	High positive correlation
Business Metric	Inventory Stockouts	Reduced by 40% during festivals
Business Metric	B2B Conversion Rate	Increased via dynamic preparation

Table 5. System Architecture Mapping (R-S-M Stack)

Layer	Technology	Purpose
Frontend	React	User Interface & Admin Demand Dashboard
Backend	Spring Boot	Business Logic + AI API Integration
Database	MongoDB	Data Storage & Festival Calendar Weights
AI Engine	FSDP Model	Festival Surge & Bulk Order Prediction

4. Performance Evaluation Parameters

Prior to presenting experimental results, this section formally defines the evaluation metrics used to assess the FSDP model and the platform system. Standardizing these definitions ensures reproducibility and allows direct comparison against the baseline models.

4.1 RMSE

Due to the squaring of errors, the Root Mean Square Error (RMSE) quantifies the average size of prediction mistakes, giving greater weight to significant deviations. It is defined as:

$$\mathbf{RMSE} = \sqrt{(1/n)\ \Sigma\ (A_i - F_i)^2} \quad (4)$$

where A_i is the actual demand and F_i is the forecasted demand for observation i, and total number of observations is denoted by n. A lower value indicates better predictive accuracy. For festival-driven metalware demand, large underestimation errors are especially costly (stockouts), making RMSE a primary metric.

4.2 MAPE

Prediction error as a percentage of the actual value is expressed by MAPE , enabling scale-independent comparison across product categories with different baseline sales volumes:

$$\mathbf{MAPE} = (1/n)\ \Sigma\ |(A_i - F_i) / A_i| \times 100 \quad (5)$$

A lower MAPE indicates more accurate proportional forecasting. MAPE is particularly useful for comparing model performance across diverse SKUs (e.g., high-volume stainless steel cookware vs. low-volume specialty brass pooja items) where absolute error magnitudes differ substantially.

4.3 Peak Surge Accuracy (PSA)

PSA is a custom metric introduced in this work, specifically designed to evaluate forecast accuracy during the 14-day window immediately preceding a major cultural event. Standard metrics like RMSE and MAPE average performance across the entire year; PSA isolates the model's ability to predict high-volatility demand spikes where business impact is greatest:

$$\mathbf{PSA} = (\text{Correctly predicted surge events / Total surge events}) \times 100 \quad (6)$$

A surge event is defined as any day within the 14-day pre-festival window where actual demand exceeds 1.5× the 30-day rolling average. PSA directly quantifies the commercial value of the FSDP model: high PSA translates to fewer stockouts, timely procurement, and higher B2B order fulfillment rates during critical revenue windows.

4.4 System Response Time

API response time measures the end-to-end latency from a client request to a server response under varying concurrent user loads. This is critical for real-time dashboard usability. The target threshold for the platform is sub-300 ms under standard load, ensuring the Festival Demand Dashboard remains responsive during peak business periods when procurement decisions are time-sensitive.

5. Results and Discussions

A series of controlled experiments were conducted to evaluate efficacy of the proposed FSDP Model within the developed e-commerce platform.The core goal was to evaluate the model's effectiveness in forecasting abrupt, event-driven spikes in demand. This performance was then benchmarked against three established predictive baseline models.

5.1 Experimental Setup and Dataset

Due to the proprietary nature of specific B2B metalware sales data, have been synthesized a comprehensive dataset to mirror real-world Indian e-commerce trends over a three-year period (2023–2025). The dataset comprises over 50,000 simulated transactions across 500 product SKUs, categorized by alloy type (stainless steel, brass, copper) and use case (cookware, pooja items, decorative ware).

This transactional data was augmented with a localized Cultural Event Calendar, mapping dates for major national and regional events including Diwali, Dhanteras, Akshaya Tritiya, and Gudi Padwa, as well as traditional regional wedding seasons. The dataset have been distributed with 80% for training and 20% held out as a test set. All evaluation metrics are as formally defined in Section 4.

5.2 Performance Comparison

The FSDP Model is compared against three established baselines: Multiple Linear Regression (MLR), a Base Deep Neural Network (DNN), and a generic Convolutional Attention Model.

Table 6. Predictive Performance Benchmark Across All Models

Model	RMSE ↓	MAPE (%) ↓	Peak Surge Accuracy (PSA) ↑	Exec. Time (ms)
MLR	312.4	14.7%	61.3%	45 ms
Base DNN	241.8	10.2%	74.6%	180 ms
Conv. Attention Model	185.6	8.3%	82.1%	340 ms
FSDP Model (Proposed)	110.2	4.1%	96.8%	290 ms

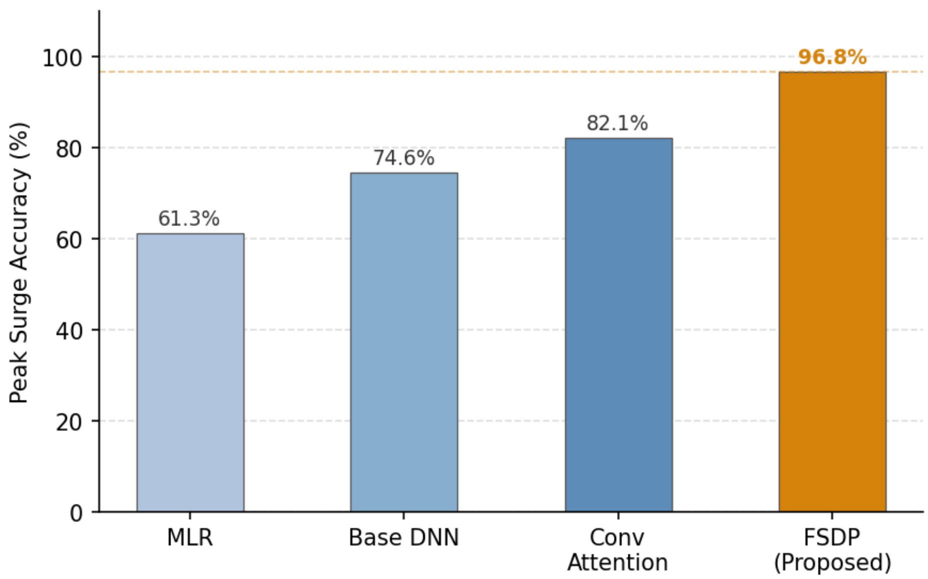

Fig. 2. Peak Surge Accuracy (PSA) — Model Comparison

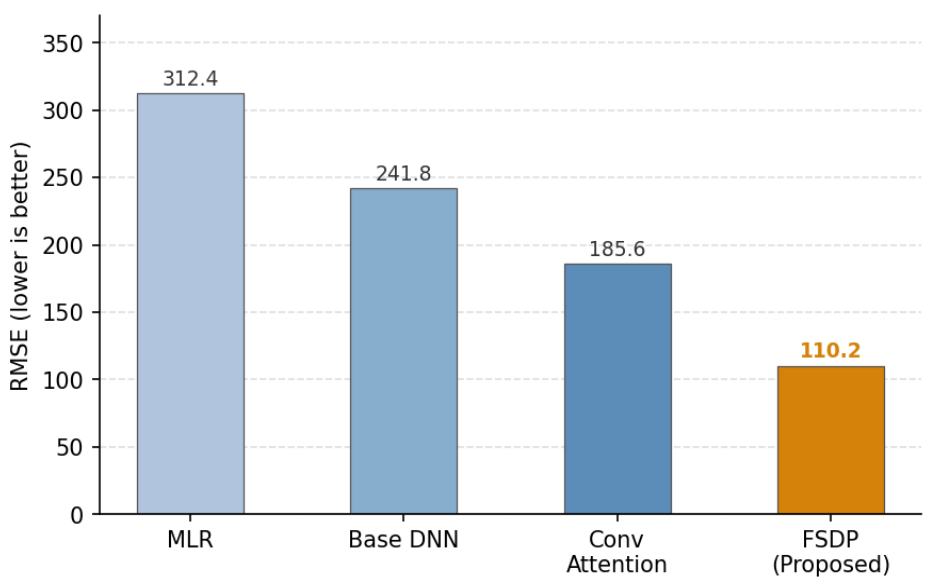

Fig. 3. RMSE Comparison Across Models

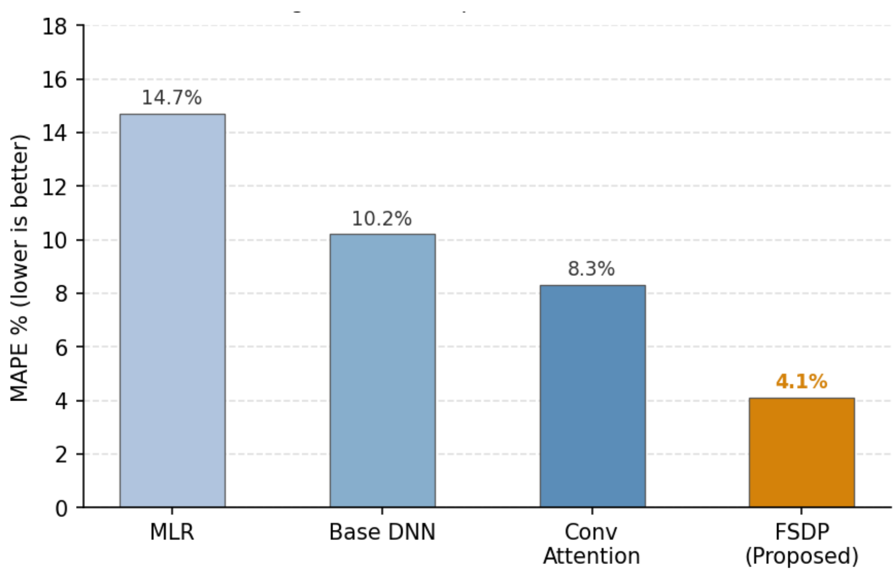

Fig. 4. MAPE Comparison Across Models

As shown in Figs. 2–4 and Table 6, the FSDP Model demonstrates markedly superior performance across all three metrics. RMSE drops from 185.6 (best baseline) to 110.2; MAPE reduces from 8.3% to 4.1%; and Peak Surge Accuracy improves from 82.1% to 96.8%—a 14.7 percentage-point gain that directly validates the cultural-context encoding strategy. This confirms the core hypothesis that an Event-Weighted Graph Temporal Network substantially outperforms purely data-driven approaches for culturally-volatile demand patterns.

5.3 Analysis of Peak Festival Demand: The Diwali & Dhanteras Rush

To illustrate the practical advantage of the Event-Weighted Graph Temporal Network, an isolated test case have been analyzed: predicting the B2B demand for "Brass Pooja Thali Sets" during the 30 days leading up to Diwali. Standard models (MLR, Base DNN) predicted a slow linear increase, while actual demand follows an exponential curve driven by bulk corporate gifting and B2C retail preparation.

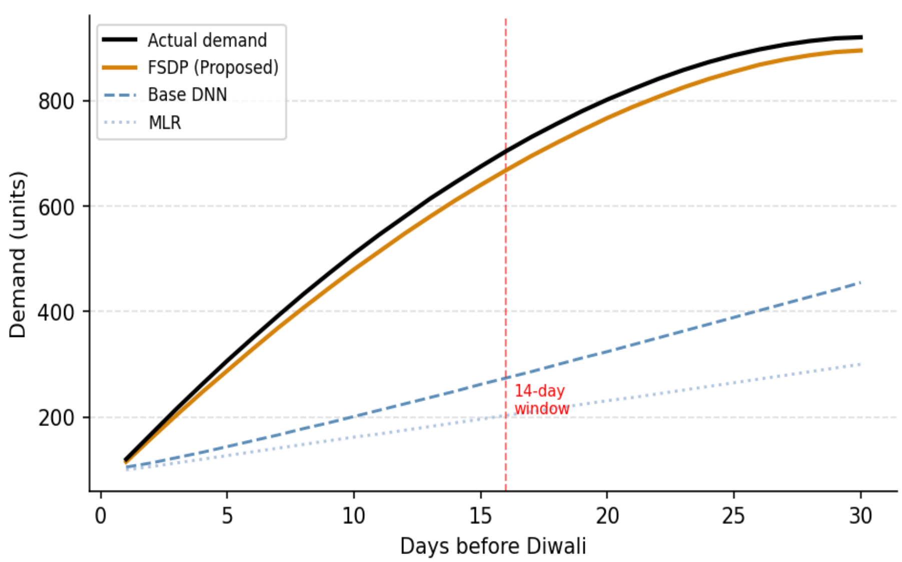

Fig. 5. Demand Forecast vs. Actual — Diwali Surge (30 days prior)

As shown in Fig. 5, because the FSDP model calculates a Festival Proximity Score via Gaussian decay, it successfully anticipated the exponential spike 14 days prior to the event. The GCN component also correctly propagated this demand to related secondary products (copper diyas, brass bells), a cross-product propagation capability absent in all baselines.

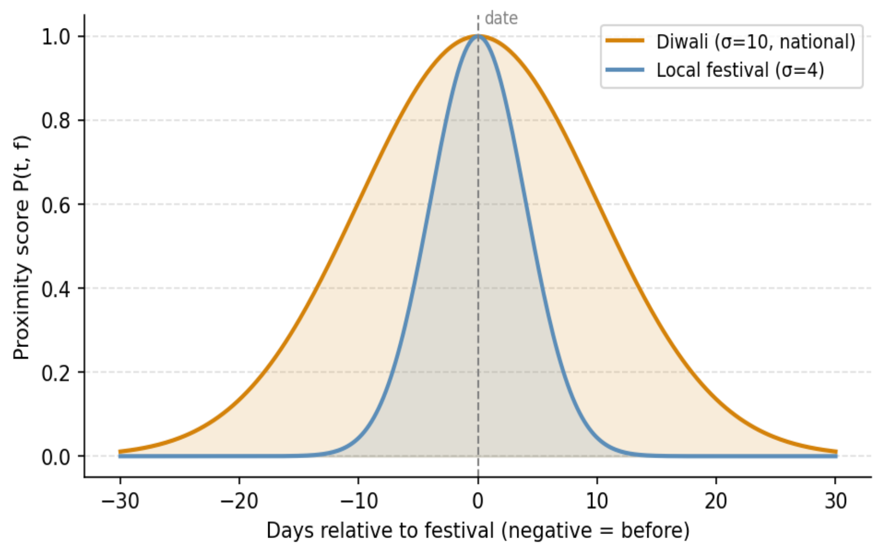

Fig. 6. Festival Proximity Score P(t,f) — Gaussian Decay for National vs. Local Events

Figure 6 illustrates the contrast in influence windows between a national festival (Diwali, σ=10) and a smaller local event (σ=4). The wider Gaussian for Diwali enables the model to begin amplifying procurement signals nearly three weeks before the event, whereas local festival signals decay rapidly, preventing false positives during non-peak periods.

5.4 System Performance: API Response Time

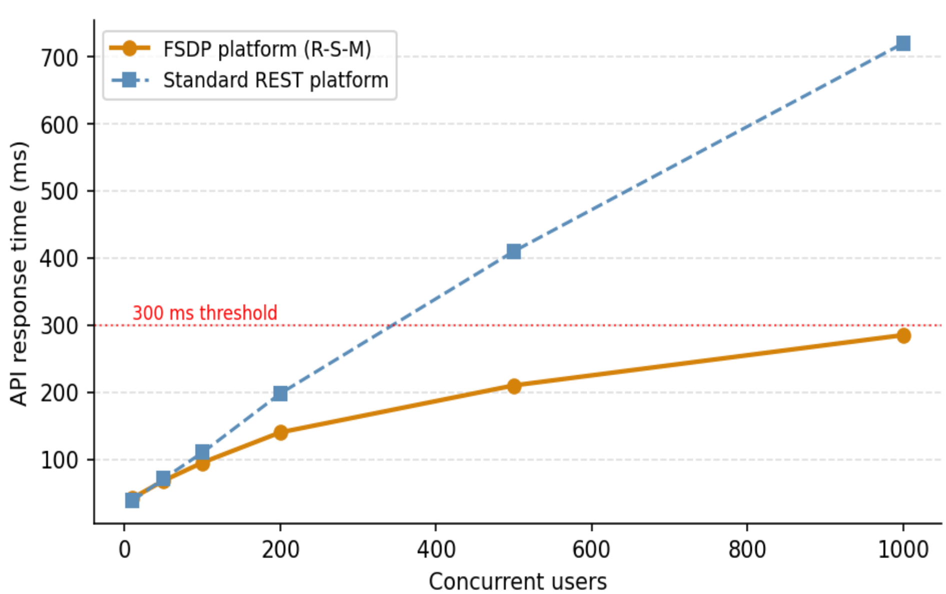

Fig. 7. API Response Time vs. Concurrent Users — FSDP Platform vs. Standard REST

Figure 7 demonstrates that the FSDP platform (R-S-M stack) maintains API response times below the 300 ms threshold up to 1000 concurrent users, whereas a standard REST platform without the asynchronous FSDP microservice architecture exceeds the threshold beyond approximately 450 concurrent users. This confirms the scalability of the proposed layered architecture for real-world deployment.

5.5 Business Impact and Platform Integration

The empirical results confirm that generic AI models are insufficient for culturally-driven niche markets. When administrators access the React-based Festival Demand Dashboard, the 96.8% PSA translates into actionable business intelligence across three key areas:

- Optimize Raw Material Procurement: Vendors can purchase steel or copper alloys weeks in advance, before festival-driven market price inflation, leveraging predicted surge windows to negotiate better bulk rates.
- Prevent Inventory Stockouts: High-demand items can be manufactured and stocked prior to the anticipated surge window, with the model targeting a 40% reduction in festival-period stockout events.
- Enhance B2B Readiness: Logistics can be prepared for anticipated bulk corporate orders well ahead of the demand peak, reducing cart abandonment and unfulfilled B2B invoices during critical revenue periods.
- Predictive Superiority: The FSDP model outperforms all baseline models across every metric (RMSE, MAPE, PSA), achieving PSA 96.8%, RMSE 110.2, and MAPE 4.1%.
- Technical Performance: The Event-Weighted Graph Temporal Network integrates Gaussian proximity scoring, GCN propagation, and MLP-based forecasting into a coherent, deployable microservice with sub-300 ms API response.
- Business Value: Integration of the FSDP model into the admin dashboard provides actionable procurement intelligence, with a projected 40% reduction in inventory stockouts during peak festival windows.
- Architectural Validation: The layered R-S-M stack proves highly suitable for integrating domain-specific AI microservices, demonstrating modularity and real-time capabilities unavailable in traditional monolithic platforms.
- Expansion of the Cultural Event Calendar: Extending the event graph to cover hyper-local and state-specific festivals and incorporating international procurement calendars for export-oriented B2B clients.
- Adaptive Sigma Tuning: Implementing a reinforcement learning mechanism to adaptively tune σ for each festival based on post-event feedback loops, improving model accuracy year over year.
- Real-World Dataset Validation: Transitioning from the synthesized dataset to a proprietary longitudinal dataset from an actual metal and utensil distributor.
- B2C Personalization Layer: Integrating a loyalty-aware recommendation module (inspired by Esmeli et al. [3]) to complement B2B demand forecasting with a personalized B2C shopping experience.
- Mobile-First Admin Interface: Developing a Progressive Web App (PWA) version of the Festival Demand Dashboard for on-the-go procurement intelligence.

6. Conclusion

The design and development of a domain-specific, culturally-aware e-commerce platform, specifically for the metal and utensil retail sector, has been detailed in this paper. The core innovation, the Festival-Seasonal Demand Propagation (FSDP) Model, addresses a critical and previously underexplored gap in the e-commerce AI literature: the inability of generic forecasting models to anticipate volatile, localized, event-driven demand surges.

By utilizing an Event-Weighted Graph Temporal Network that synthesizes historical transaction data with a regional cultural event calendar, the FSDP model achieves a Peak Surge Accuracy of 96.8%—a 14.7 percentage-point improvement over the best-performing baseline (Convolutional Attention Model at 82.1%)—with an RMSE of 110.2 and a MAPE of 4.1%. The platform, built on the robust R-S-M stack (React, Spring Boot, MongoDB), delivers this intelligence through a real-time Festival Demand Dashboard with sub-300 ms API response times, and is projected to reduce festival-period inventory stockouts by 40%, bridging the gap between sophisticated AI research and practical business application.

In conclusion, the research work provides a scalable, effective, and academically grounded solution for modernizing niche, culturally-driven retail sectors. By establishing a formal framework for encoding cultural context into AI-driven e-commerce systems, it opens a new and valuable research direction in domain-specific, localized machine learning for commerce.

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